Concept of Health & Disease

ASS. PROF. DR. TASNEEM M. ALI

PUBLIC HEALTH & COMMUNITY MEDICINE

SOHAG & MERIT UNIVESITIES

LEARNING OBJECTIVES

- Define health and describe its dimensions
- Recognize the spectrum of health
- Define well-being and its components
- Understand determinants of health
- Explain disease concept and epidemiological triad
- Describe natural history and stages of disease

DEFINITION OF HEALTH (WHO)

The Definition People Often Miss....

What is HEALTH?

Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity (World Health Organization, WHO)

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TYPES OF DIMENSIONS OF HEALTH

- physical dimension
- mental dimension
- social dimension
- spiritual dimension
- emotional dimension
- vocational dimension

PHYSICAL DIMENSION

- It conceptualizes health, biologically as a state in which every cell and every organ functioning at optimum capacity and in perfect harmony with rest of the body.
- Signs of physical health in the body are-

A) Good complexion

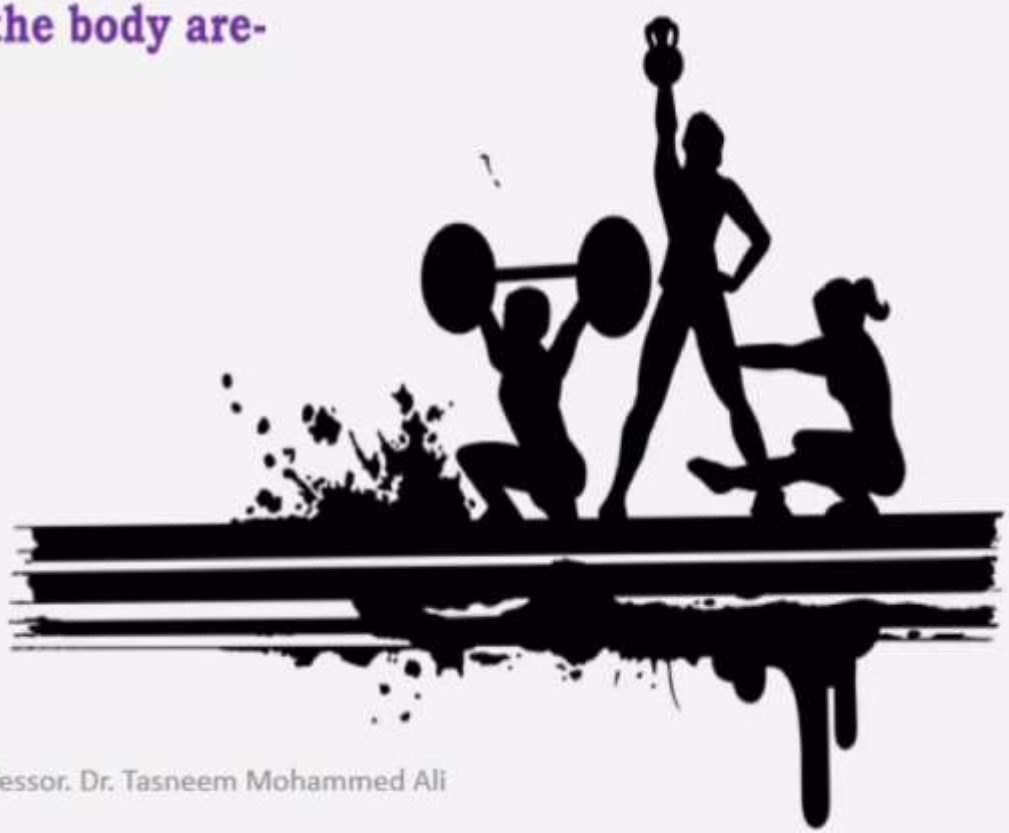
B) Clean skin

C) Bright eyes

D) Sweet breath

E) Good appetite

F) Sound sleep etc.



PHYSICAL DIMENSION OF HEALTH

Normal functioning of organs and body systems

clinical examination

Medical history and hospitalization

Anthropometric measurements

Laboratory and biochemical investigations

MENTAL DIMENSIONS

- A good mental health is defined as the " state of balance between the individual and the surrounding world, a state of harmony between self and others."
- The scientific foundations of mental health are not yet clear leading to not have precise tools for evaluation.
- Attributes of mentally healthy person-

A) Free from internal conflicts.

B) Well adjusted

C) Searches for identity

D) Strong sense of self esteem

E) Knows himself, his needs, problems and goals

F) Good self control

G) Faces problems and tries to solve them intelligently.

MENTAL DIMENSION OF HEALTH

Not merely absence of mental illness

Ability to think clearly and coherently

Balance between individual and environment

Assessed by behavior and attitude

SOCIAL DIMENSION

- It is defined as " the quantity and quality of an individual's interpersonal ties and the extent of involvement with the community".

- It includes –

A) Level of social skill one possesses

B) Social functioning

C) Ability to see oneself as a member of a larger society that is the world

SOCIAL DIMENSION OF HEALTH

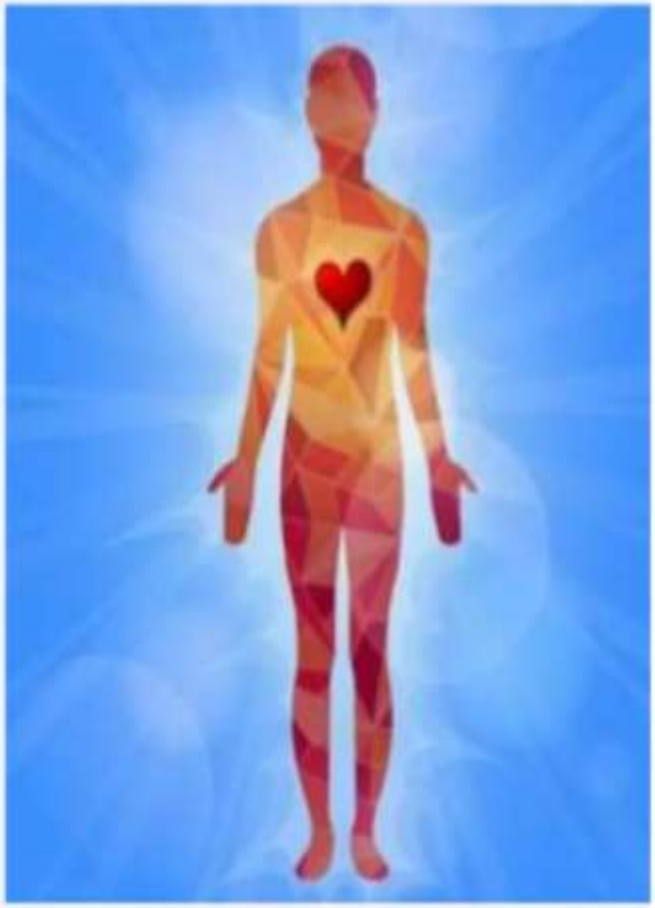
Ability to form relationships

Harmony and integration with society

Social functioning and role performance

SPIRITUAL DIMENSION

- It refers to that part of the individual which reaches out and strives for meaning and purpose for life.
- Meditations, prayers or spiritual gatherings are organized to maintain spiritual health.
- It includes love, charity, purpose, principles, ethics, integrity and hope of life.



SPIRITUAL DIMENSION

Holistic wellbeing

Sense of meaning and philosophy of life

Transcends physiology and psychology

EMOTIONAL DIMENSION

- Emotional health relates to “feelings”.

- Emotional health includes-

A) They have positive thinking and are capable of coping and adjusting self.

B) Participates in all the activities which are related to personal growth and his self esteem.

C) Ability to express feelings freely and manage feelings effectively.

D) Aware of and accept a wide range of feelings in themselves and others.

EMOTIONAL DIMENSION

Self control

Stress management

Emotional stability

VOCATIONAL DIMENSION

- When work is fully adapted to human goals, capacities, work often plays a role in promoting both physical and mental health.

- Vocational health can be evaluated by-

A) Assessing the satisfaction level at job

B) Facilities attached to job

C) Behavior of management, administrator and colleagues at job.

Vocational Dimension

Ability to work productively

Earning sufficient income

Job satisfaction

Dimensions of Health

Social Health	Emotional Health	Mental Health	Spiritual Health	Vocational Health
Ability to make and maintain relationships; harmony with society	Ability to handle emotions, control temper, and avoid tension	Clear, coherent thinking; balance between self an surroundings	"Sound mind in a sound body'; high-level thinking, values, purpose, simple life	Ability to work productively and earn suffcienty to lead life successfuily
Example	**Example**	**Example**	**Example**	**Example**
A student who collaborates well ingroup assignmen and maintains friendships	A person stayng calm instead of shoutlng when frustrated	Someone solving a problem rationally under pressure	A person who practices refiection, gratitude, and lives according to strong values	A nurse who enjoys her work and performs her dutles effectively

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GUESS
WHAT?

Definition and Skills: The ability to form and maintain healthy relationships using communication, empathy, and cooperation skills.

Key Components: Emotional readiness, resilience, and adaptability are essential components enabling effective social engagement and integration.

Influencing Factors

Cultural norms, socioeconomic status, social support networks, and community structures

Practical Examples:

Thriving students, positive workplaces, and active community participation

Definition : is the ability to recognize, manage, and express emotions constructively, vital for everyday functioning.

Key Characteristics :healthy individuals regulate reactions, cope with stress, and show resilience, patience, and stability.

Determinants and Influences

influenced by coping mechanisms, family, culture, genetics, and supportive environments.

Practical Impact

Good health improves decision-making, communication, productivity, and overall life satisfaction.

Core Functions: involves clear thinking, problem-solving, and maintaining psychological balance for effective functioning.

Determinants Both biological factors like genetics and environmental influences such as education and social support

Impact: Good health supports emotional balance, social communication, creativity, and overall life satisfaction.

Practical Applications: People use logical strategies and maintain focus during challenges to demonstrate positive health.

- **Definition** :involves purpose, meaning, values, and inner peace beyond physical and mental health.
- **Determinants** Cultural upbringing, personal experiences, philosophy, and reflective practices shape its well-being.
- **Impact on Resilience and Stability**
- fosters resilience, hope, emotional stability, acceptance, and ethical behavior in challenges.
- **Practical Practices**
- Meditation, gratitude, journaling, and moral frameworks enhance its alignment and harmony.

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- **Definition** Is the ability to work productively and find meaning while balancing work and personal life.
- **Determinants :**

Factors include employment conditions, workplace safety, income, growth opportunities, and job control.

- **Effects on Well-Being**

boosts motivation, competence, and reduces stress for overall well-being.

- **Practical Example**
- A nurse feeling valued and performing efficiently

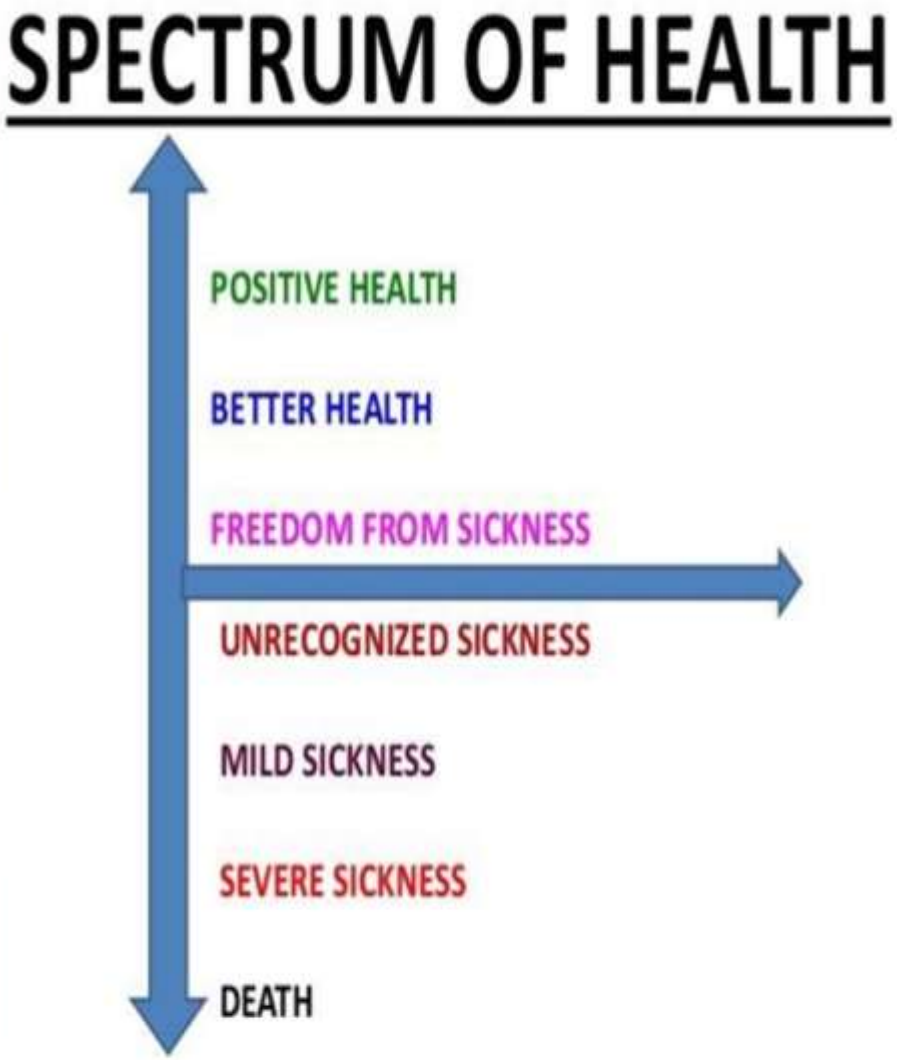
Spectrum of Health

Health and disease exist on a continuum

No clear cut boundary

Dynamic and changing process

Ranges from optimal wellbeing to death



Optimal Health → Mild Dysfunction → Severe Disease → Death

CONCEPT OF WELL-BEING

Well-being includes the presence of positive emotions and moods (e.g., contentment, happiness), the absence of negative emotions satisfaction with life, fulfillment and positive functioning. In simple terms, well-being can be described as judging life positively

Positive emotions and life satisfaction

Absence of negative emotions (e.g., depression, anxiety),

Positive functioning

Judging life positively and feeling good.

COMPONENTS OF WELL-BEING

- Physical
- Economic
- Social
- Emotional
- Psychological
- Life satisfaction
- Development and activity

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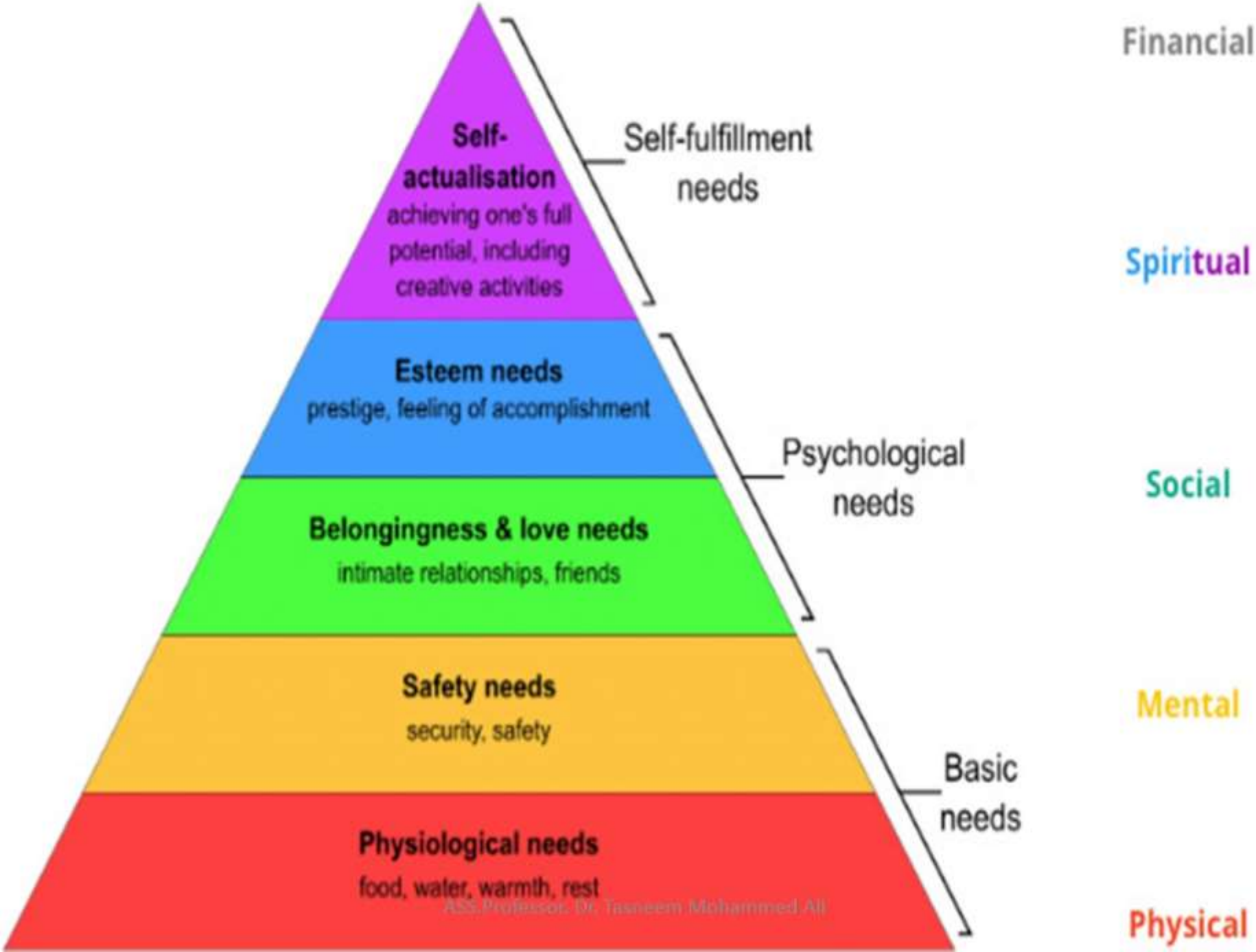

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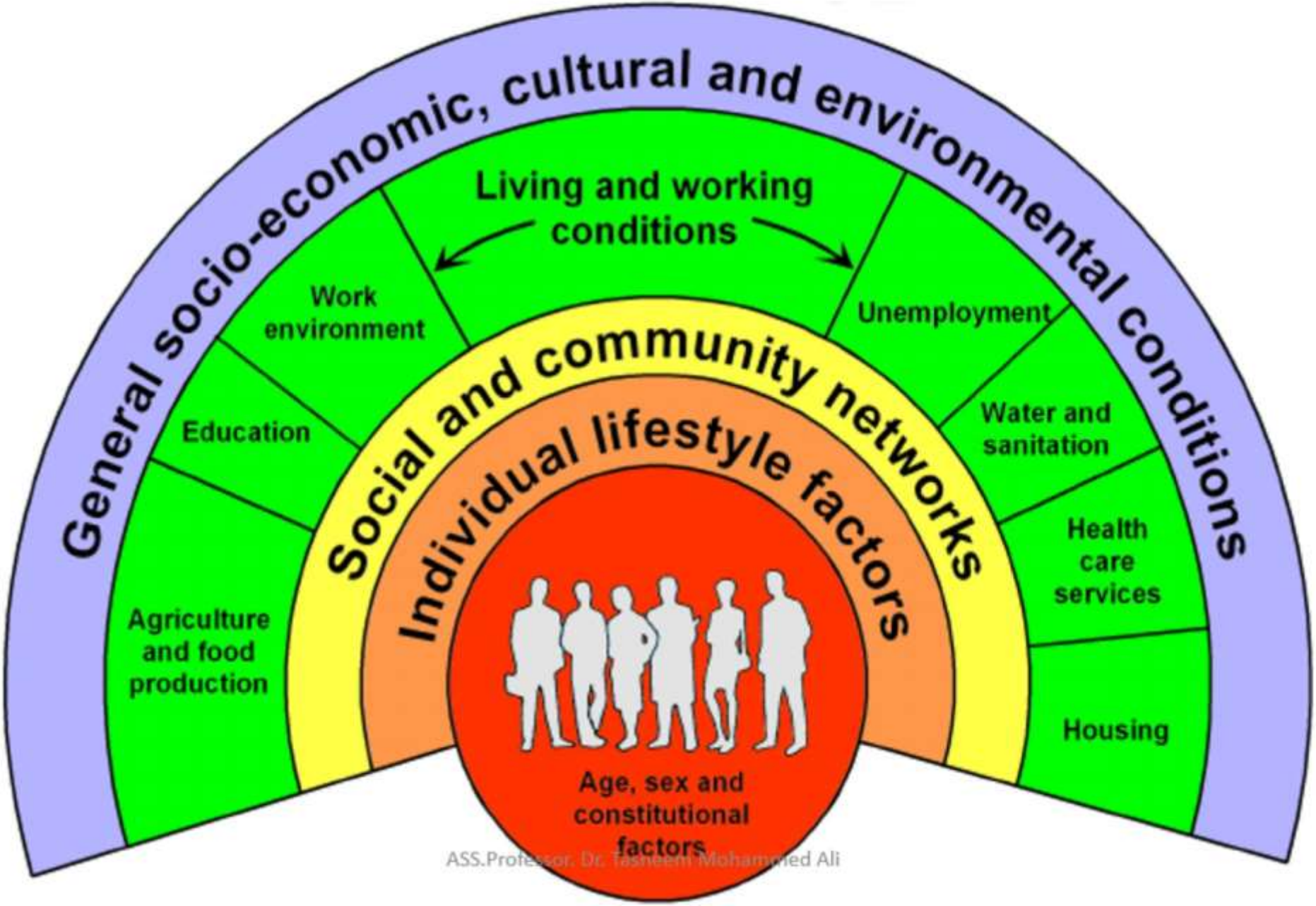
Determinants of Health



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Determinants of Health

Social and economic environment

Physical environment

Individual characteristics and behavior

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Key Determinants of Health

- Income and social status
- Education
- Physical environment
- Employment
- Social support
- Culture
- Genetics
- Health services
- Gender

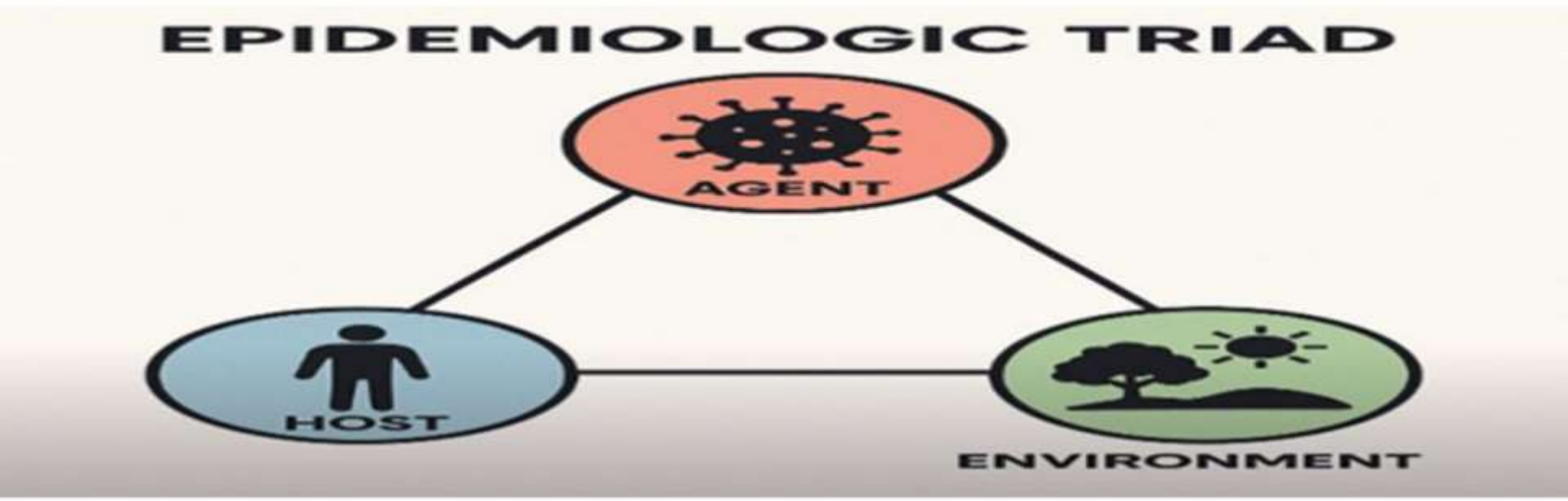
Asst. Professor, Dr. Tasnim Mohammed Ali

CONCEPT OF DISEASE

Disease :A condition, in which body health is impaired and performance of vital functions in the body is interrupted.' disease is a physiological or psychological dysfunction of the body

Illness :a subjective feeling of not being well

Sickness :state of social dysfunction (i.e. inability to perform his 'Social role').



- **The agent factors:** Related to the various characteristics of the "agent" which causes the disease.
- **The host factors:** Which relate to various characteristics of the human being himself.
- **Environmental factors:** Which relate to the various characteristics of the environment in which the human being is living.

EPIDEMIOLOGICAL TRIAD

- ***Agent - Host - Environment interaction causes disease***

Disease Agent

- **Survivability**
  Of disease agent in its own environment *outside* of host
- **Infectivity**
  Ability of a disease agent to enter, survive in, and multiply and causing disease in a susceptible host. (measles)
- **Virulence**
  The capacity of a agent/pathogen to cause severe, widespread disease in host.

Host's Environment

- Weather
- Shelter/housing
- Geography/location
- Occupational/job setting, etc.
- Air quality
- Food, water avail.
- Sanitation, etc.

Host

- Age
- Health (immune status, etc.)
- Gender
- Genetic predisposition
- Behavior (of host)
- Nutritional, hydration, sleep status

host [hōst] 1. *any animal/person/plant that harbors and provides sustenance for another organism (a parasite, disease pathogen, etc.) – whether "sick" or not*

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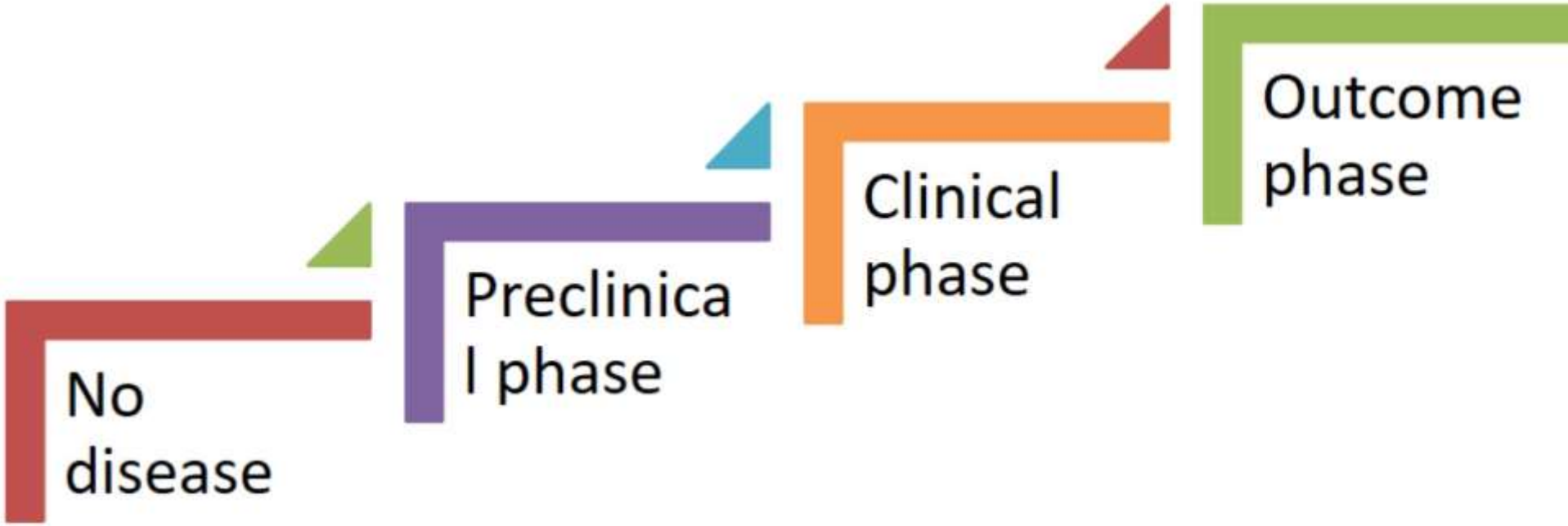
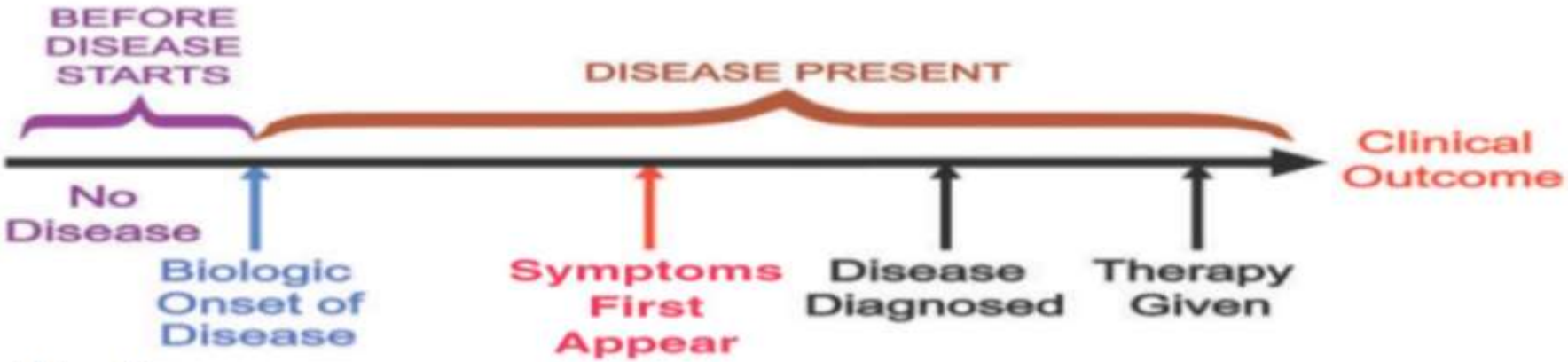
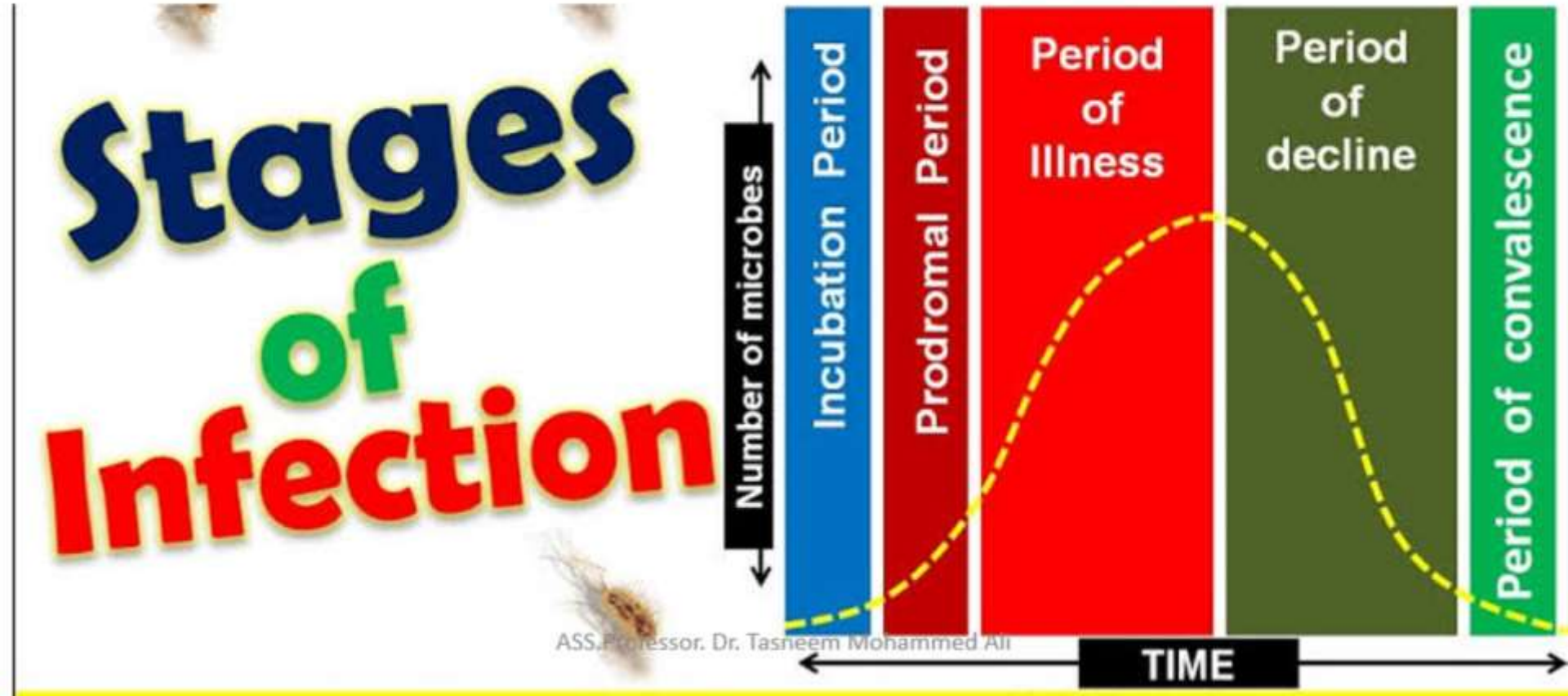
NATURAL HISTORY OF DISEASE



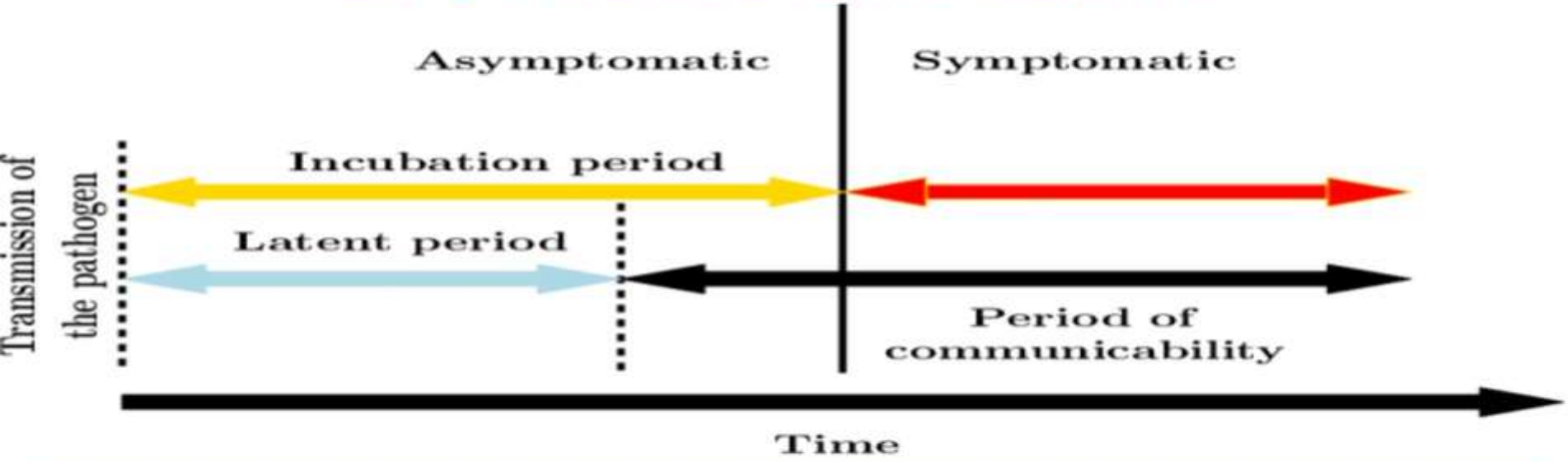


- **No disease**
- **Preclinical phase** (biological onset of disease = pathological phase)
- **Clinical phase**
  - Symptoms (individual feeling), Signs (physician diagnoses)
  - Treatment (physician recommends treatment, such as drugs, surgery, or other therapies)
- **Outcome phase**
  - Complete recovery (cure), Recovery but with remaining pathology or handicap, Death

STAGES OF INFECTIOUS DISEASE



INCUBATION PERIOD



The period between the successful entrance of the organism and the onset of the first

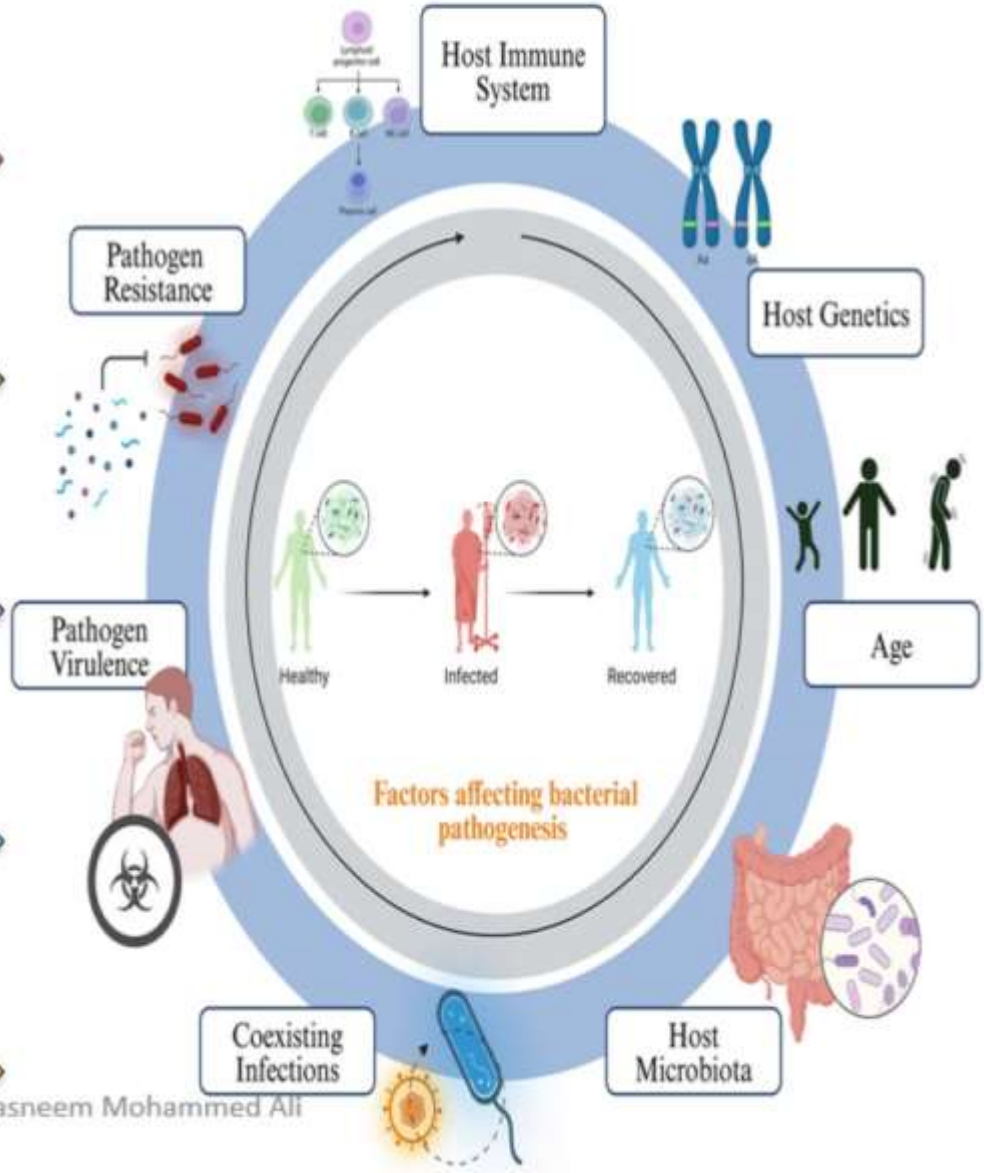
can vary from a day or two in acute disease to months or years in chronic disease, depending upon the pathogen.

Patient unaware of disease

Depends on pathogen and host immunity

Factors involved in determining the length of the incubation period

- strength of the pathogen
- strength of the host immune defenses
- site of infection
- type of infection
- the infectious dose received.



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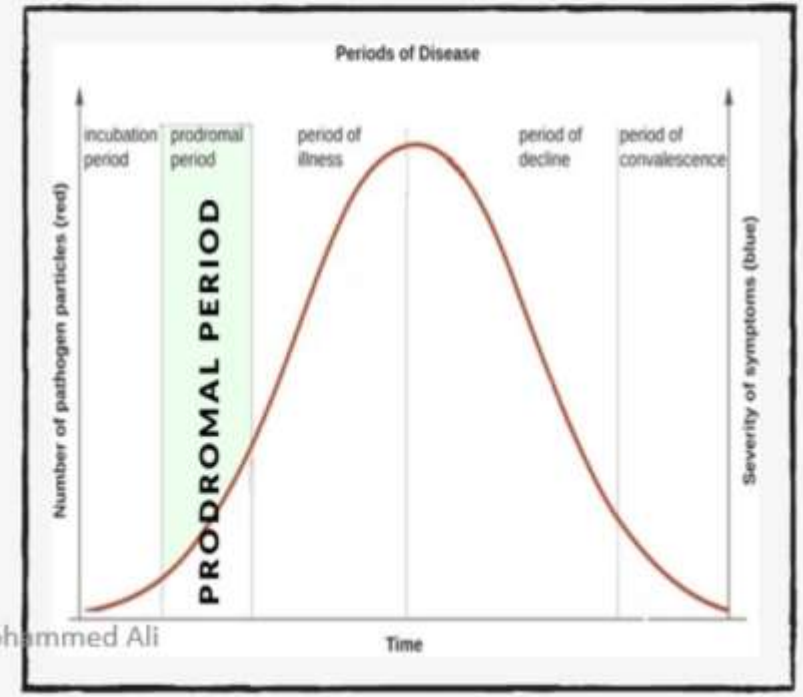
Prodromal period

Prodromal period is the period between the incubation and the illness periods of a disease.



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PRODROMAL STAGE

Early general symptoms

Non specific signs

Immune system activation

Occurs after the incubation period.

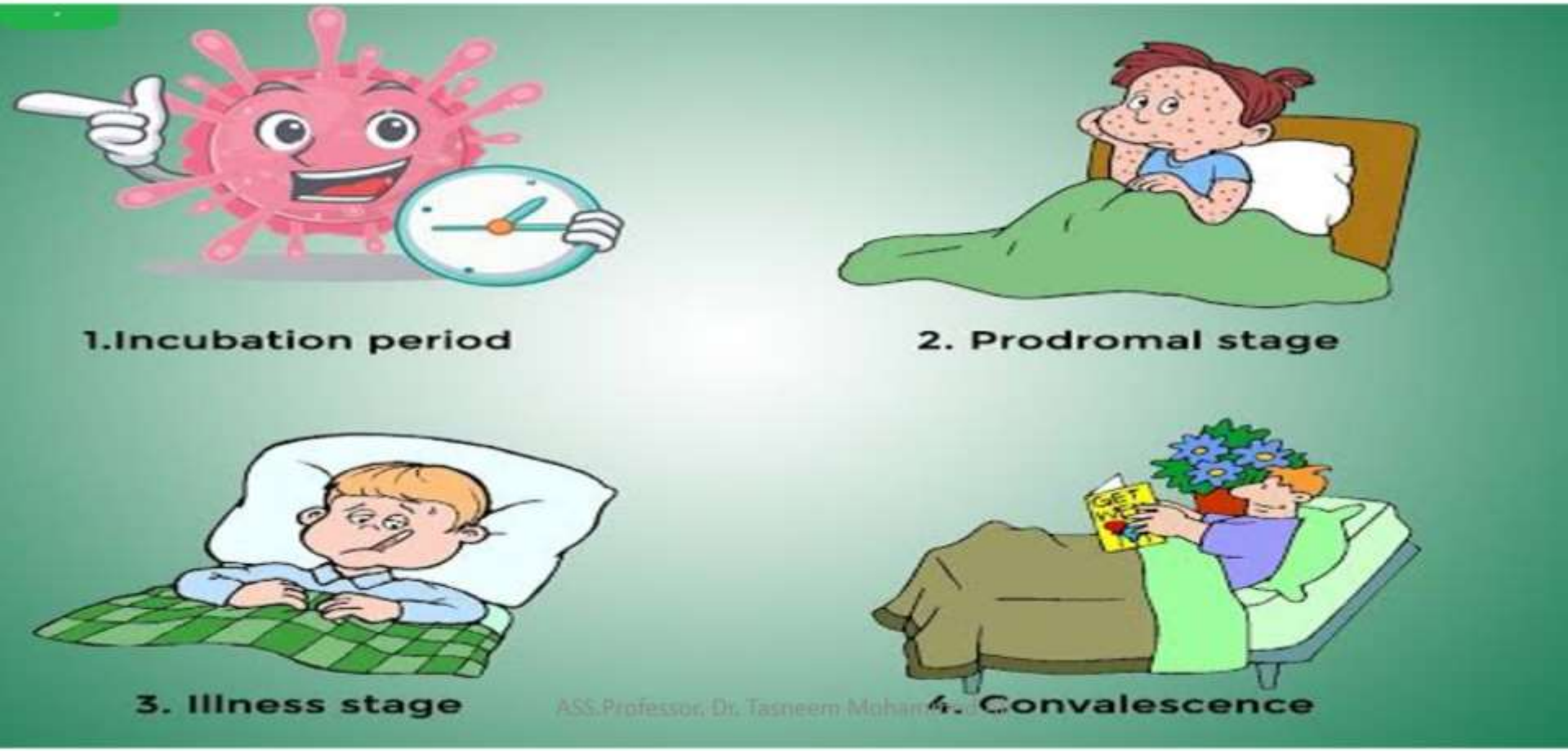
the pathogen continues to multiply and the host begins to experience general signs and symptoms of illness, which typically result from activation of the immune system,

such as fever, pain, soreness, swelling, or inflammation. Usually, such signs and symptoms are too general to indicate a particular disease

Illness Stage

Clear signs and symptoms

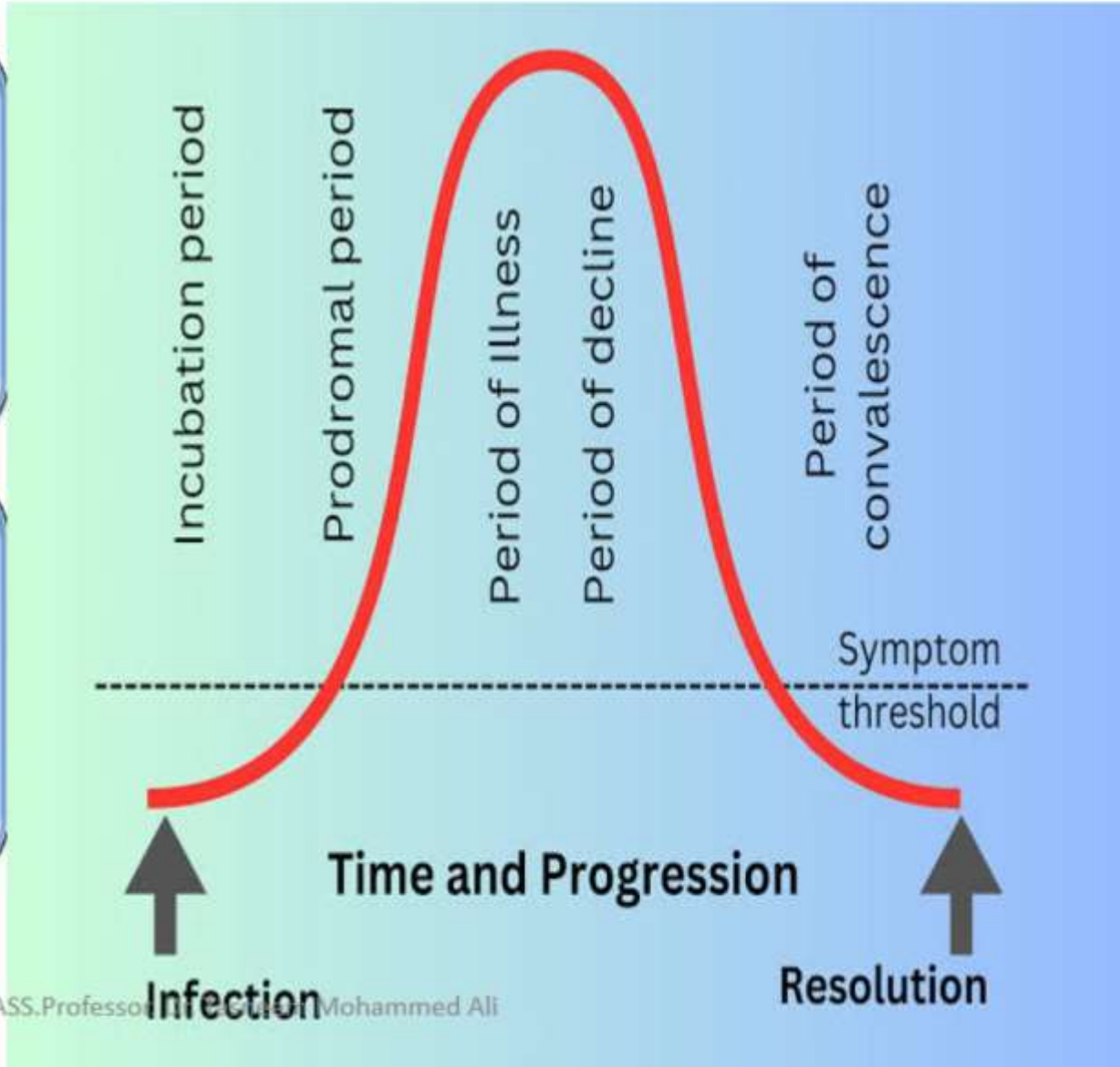
Disease most severe



DECLINE STAGE

Symptoms decrease

Risk of secondary infections



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CONVALESCENCE STAGE

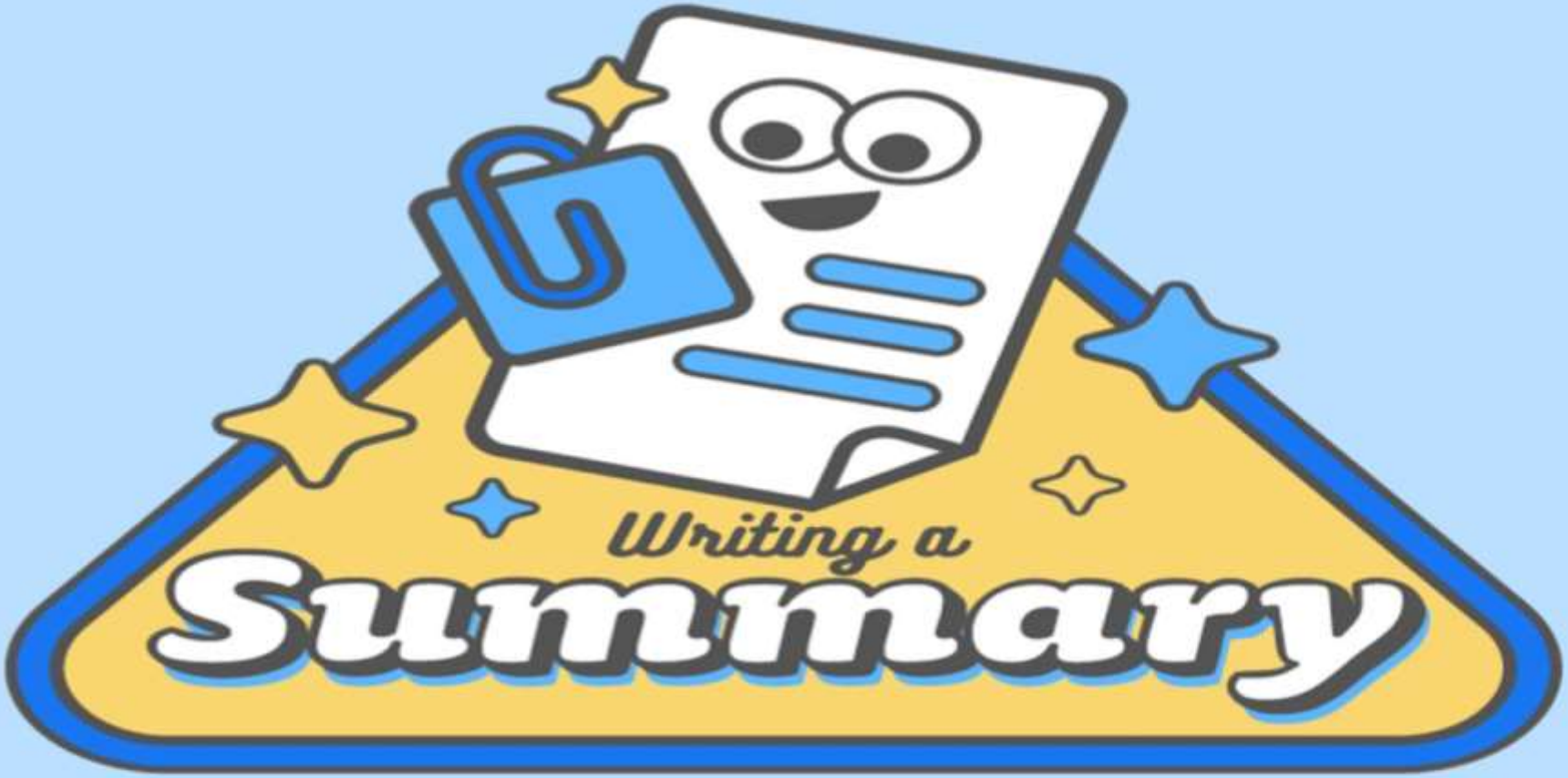
Recovery phase

Return to normal function

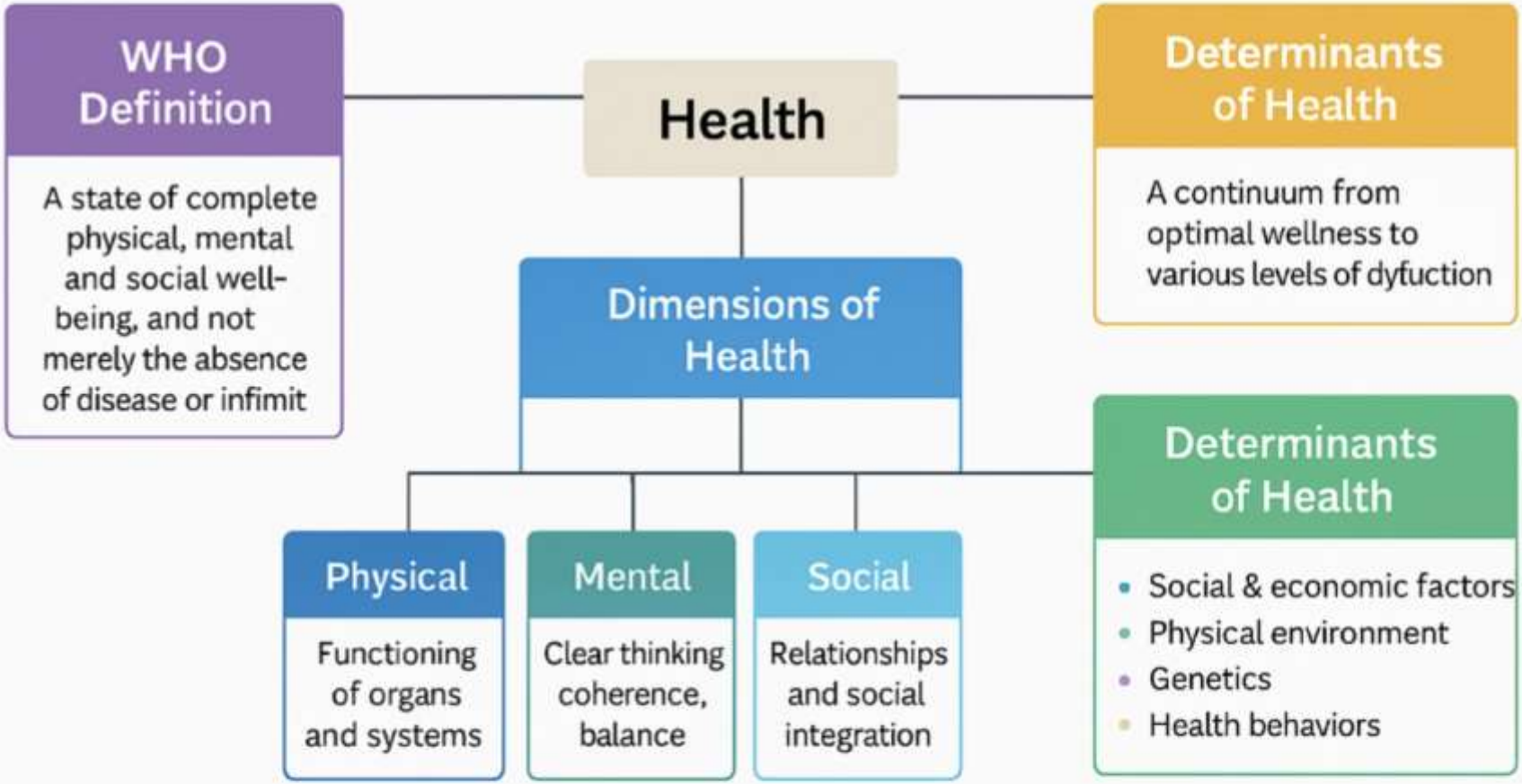
Possible permanent damage

INFECTION VS. DISEASE

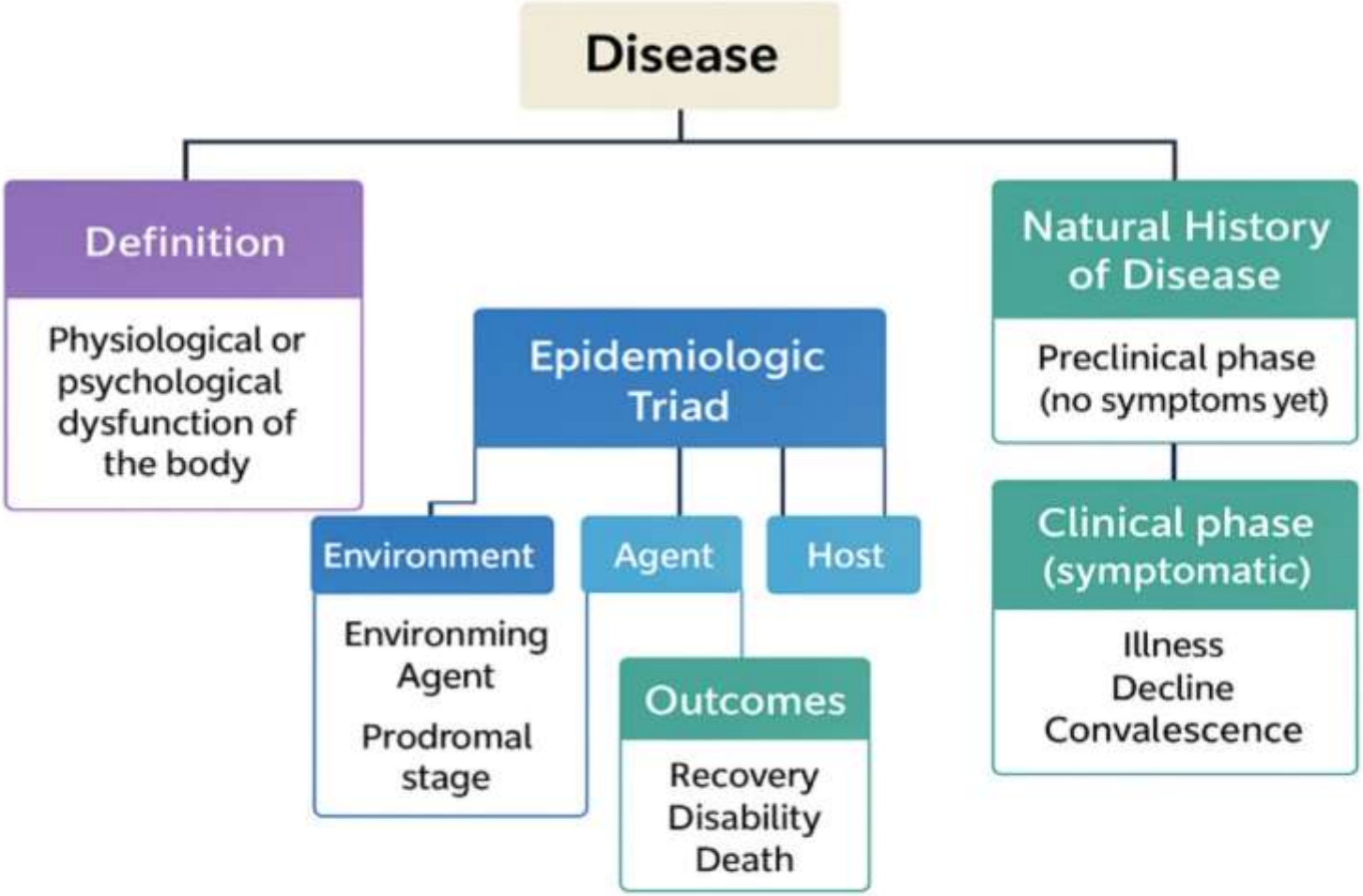
- Infection occurs when a pathogen (like a virus, bacterium, fungus, or prion) enters and multiplies within a host.
- Disease happens when these pathogens damage body cells, tissues, or organs, resulting in abnormal function and clinical symptoms.
- Our immune system fights infections but can sometimes cause collateral damage to our own cells, contributing to disease symptoms.
- Not all diseases are caused by infections; examples include:
- Diabetes
- Heart disease
- Cancer


Writing a
Summary

Concept of Health



Concept of Disease



Checking for Understanding

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Which statement best reflects the dynamic nature of the health spectrum?

A. Health is a fixed state determined only by genetics.

B. Individuals move continuously between different levels of wellness and illness.

C. Health and disease are two separate unrelated entities.

D. The transition from health to illness is always abrupt.

Which individual demonstrates optimal social health?

A. A person with no disease but constant anxiety.
B. A person who maintains satisfying relationships and integrates well with society.
C. A person with no hospital admissions in the past year.
D. A person who meditates daily.

Which example best illustrates emotional health rather than mental health?

- A. Solving complex problems logically.
- B. Maintaining calm and self-control during stressful situations.
- C. Having high-level thinking and life philosophy.
- D. Absence of psychological disorders.

A community worker wants to improve population health by targeting the most powerful non-medical determinants. Which area should they prioritize?

- A. Hospital construction
- B. Income and social status
- C. Genetic screening
- D. Improving diagnostic technology

Which scenario best describes subjective well-being?

A. Blood pressure measurement
B. Clinical diagnosis of illness
C. A person reporting life satisfaction and happiness
D. A physician noting physical signs

A factory worker exposed to toxic chemicals develops illness. According to the epidemiological triad, which component is represented by the chemicals?

A. Host
B. Agent
C. Environment
D. Vector

Which period of disease is most dangerous for silent community transmission?

A. Convalescence
B. Illness period
C. Incubation period
D. D. Decline period

A patient states they feel ill although no pathology is detectable. This situation represents

A. Disease
B. Illness
C. Sickness
D. Recovery

Which determinant explains why well-educated individuals often have better health outcomes?

A. They have stronger genetics.
B. Education reduces stress and increases self-confidence.
C. Education eliminates environmental hazards.
D. Education directly increases immunity.

A patient recovering from infection starts feeling better but becomes vulnerable to secondary infections. Which disease stage is this?

A. Incubation period
B. Prodromal stage
C. Illness stage
D. Decline period

Thank
you!